Technical Report
RAG-Based Product Details Assistant
Using Streamlit and Snowflake Cortex Search

Version 1.0

Product Details Search

Created By : Alok Kumar
Date : 13Oct'26

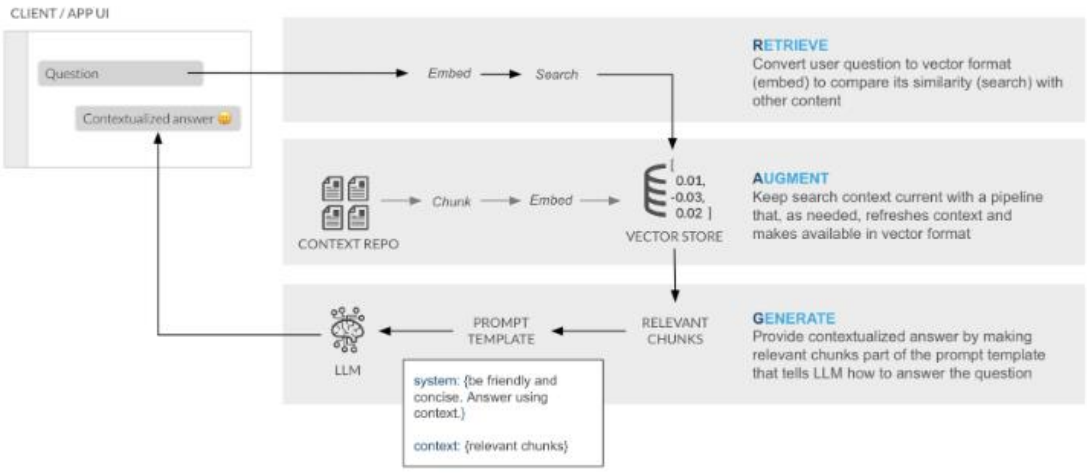
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1. Overview

This solution is a Retrieval Augmented Generation (RAG) assistant developed using Streamlit as the user interface and Snowflake Cortex Search as the retrieval engine. The application enables users to search product specification documents, retrieve contextually relevant content, and generate AI-powered responses and insights. PDF documents uploaded to Snowflake stage are indexed automatically and made searchable through Cortex Search.

2. Architecture

The architecture consists of four layers: Document Ingestion Layer (PDFs stored in Snowflake stage), Indexing and Retrieval Layer (Snowflake Cortex Search), Generative AI Layer (Snowflake Cortex LLM models such as Claude Sonnet), and Presentation Layer (Streamlit UI for chat, document library, and insights).

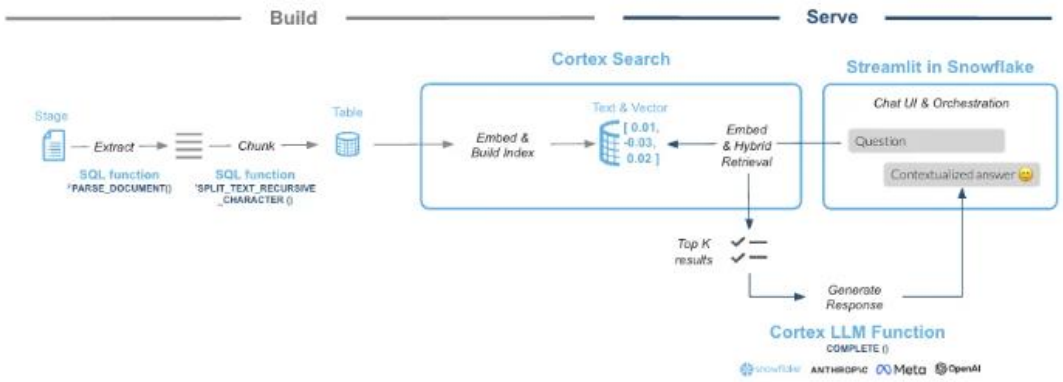


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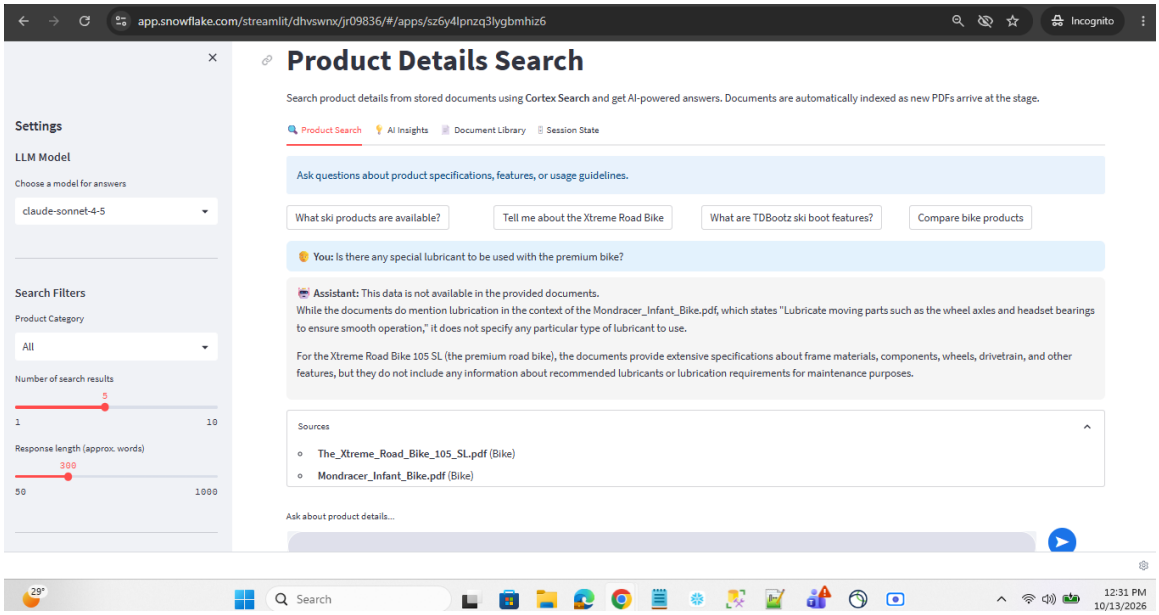
3. Flow Diagram

User Query -> Streamlit UI -> Cortex Search -> Relevant Chunks Retrieval -> LLM Context Augmentation -> Generated Response -> Display to User.



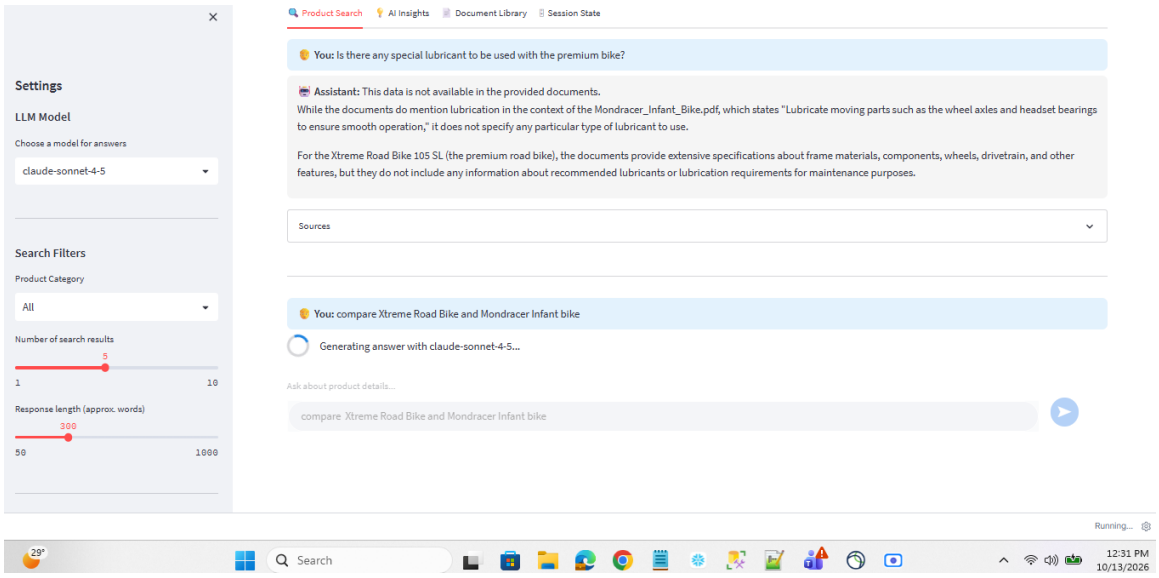
4. Output

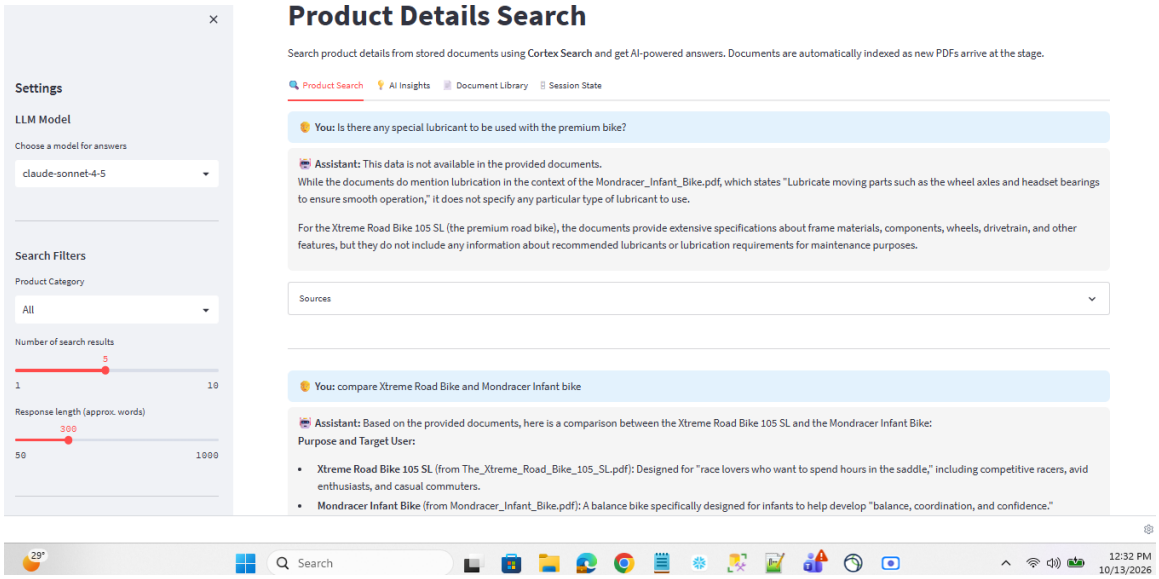
The assistant provides product-specific answers, document summaries, sentiment analysis, classification, raw document chunks, document library browsing, and conversational search capabilities.



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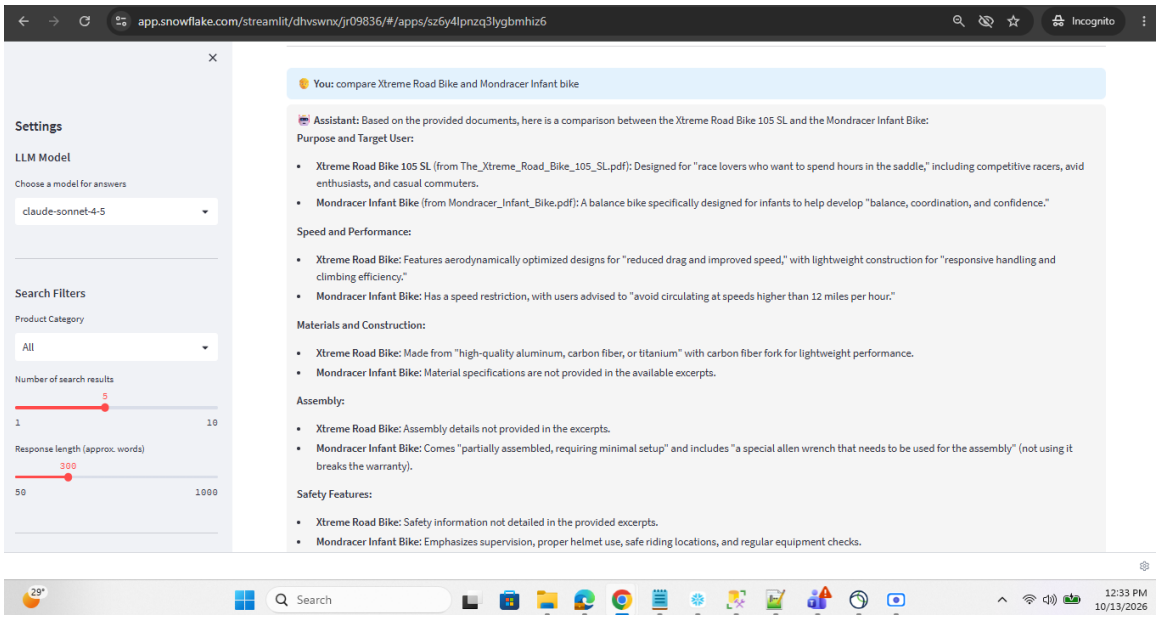
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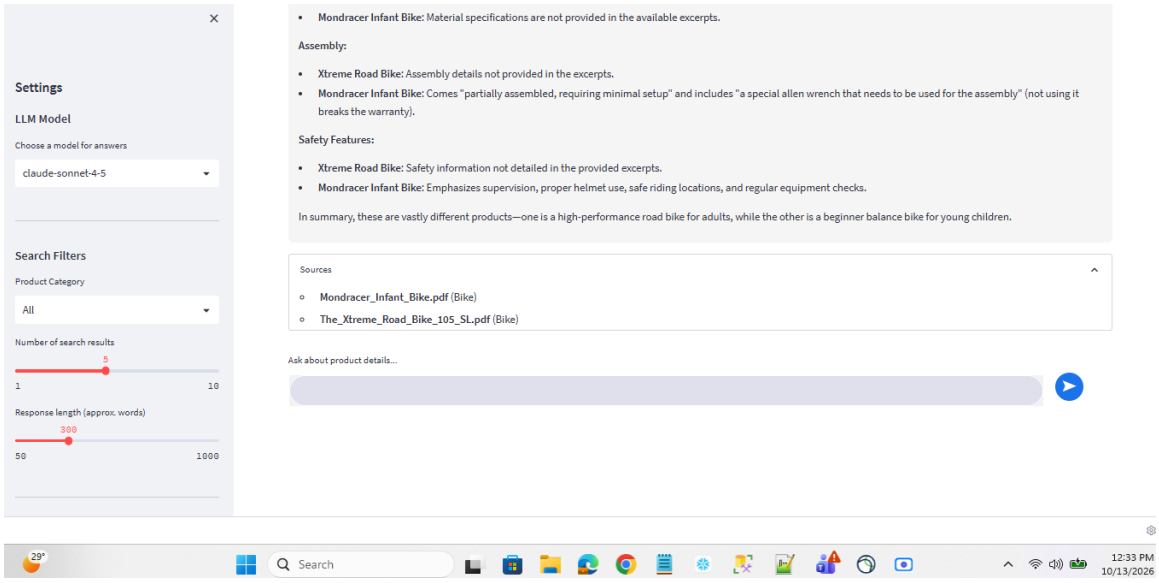




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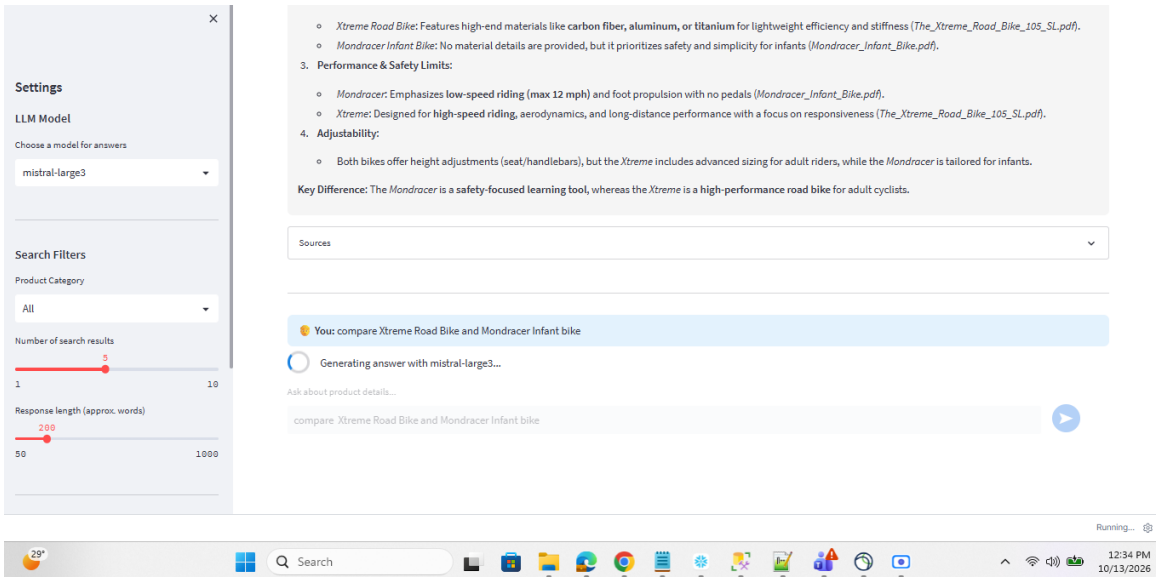
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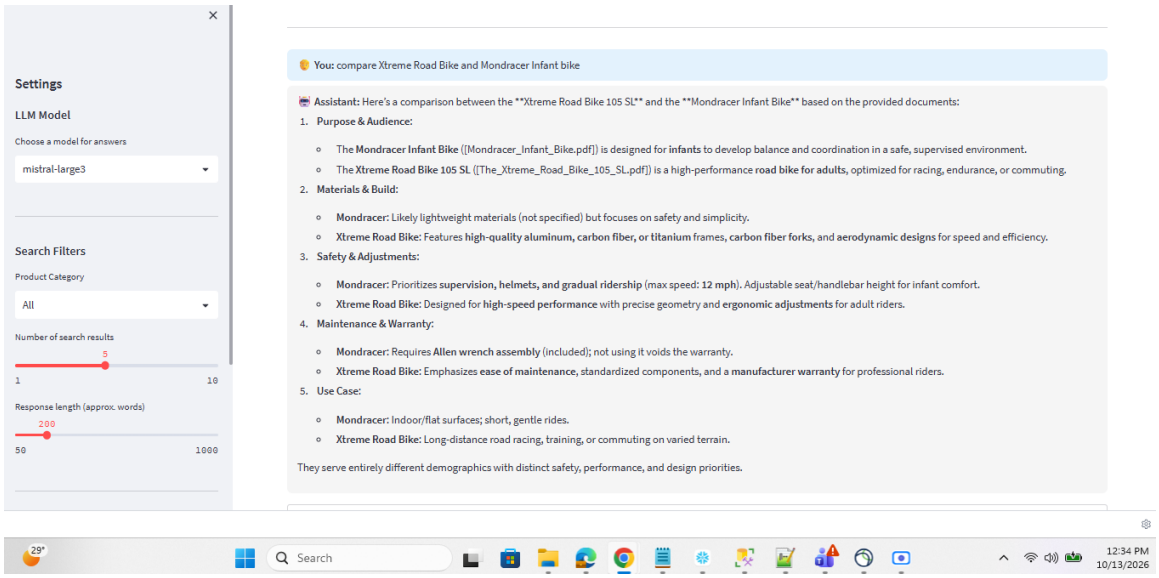




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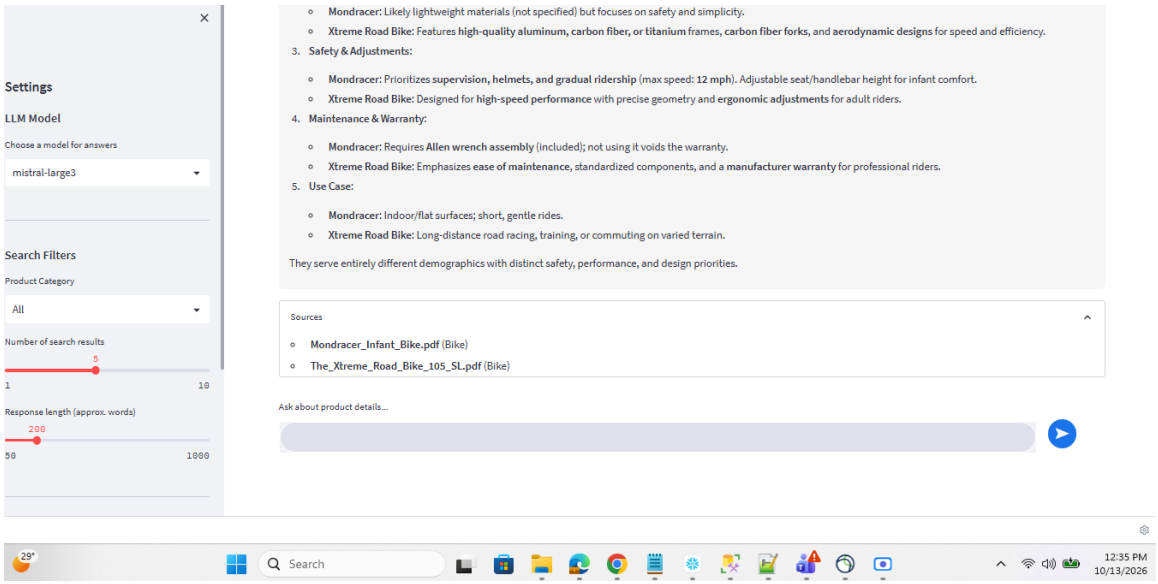
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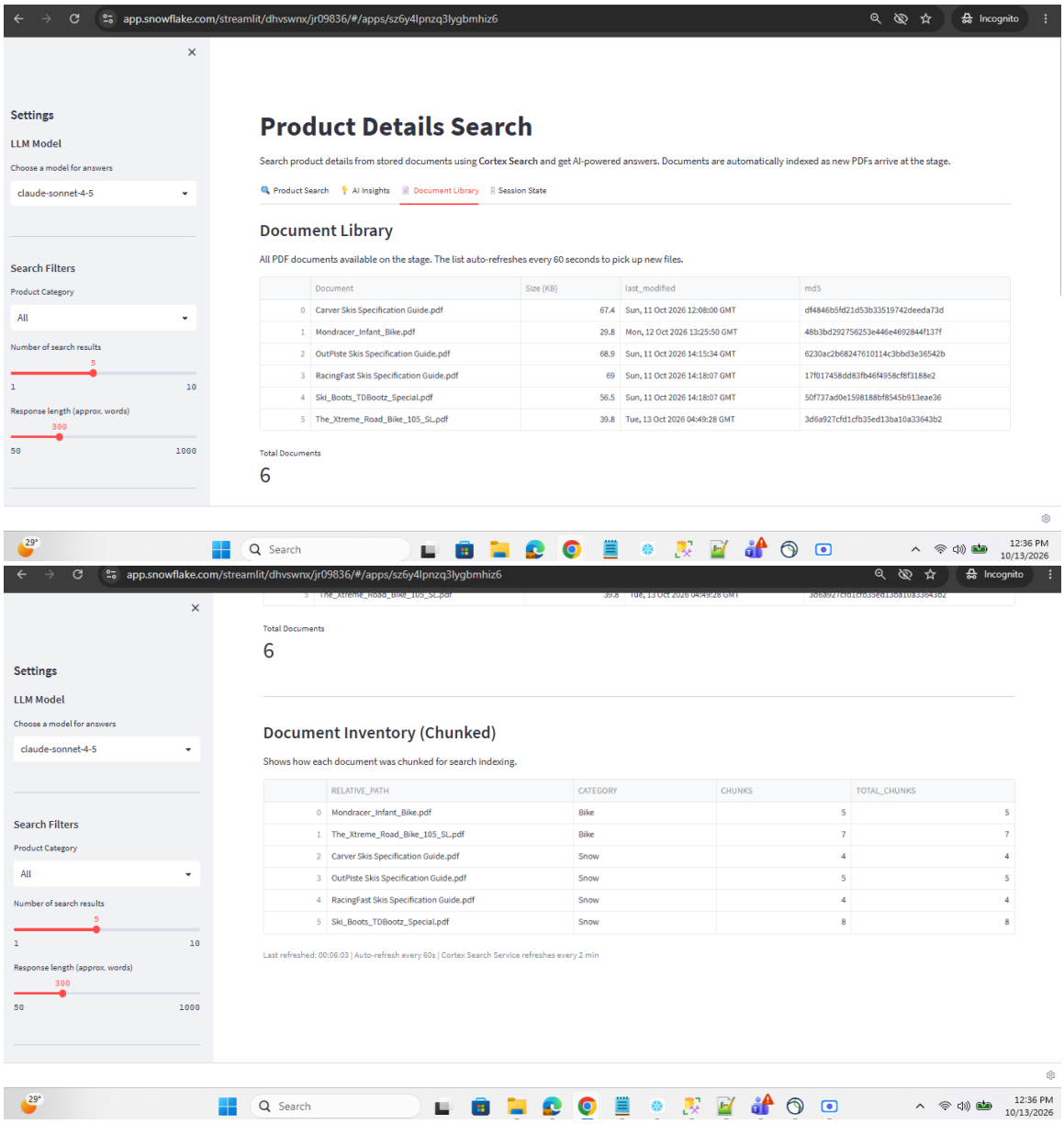
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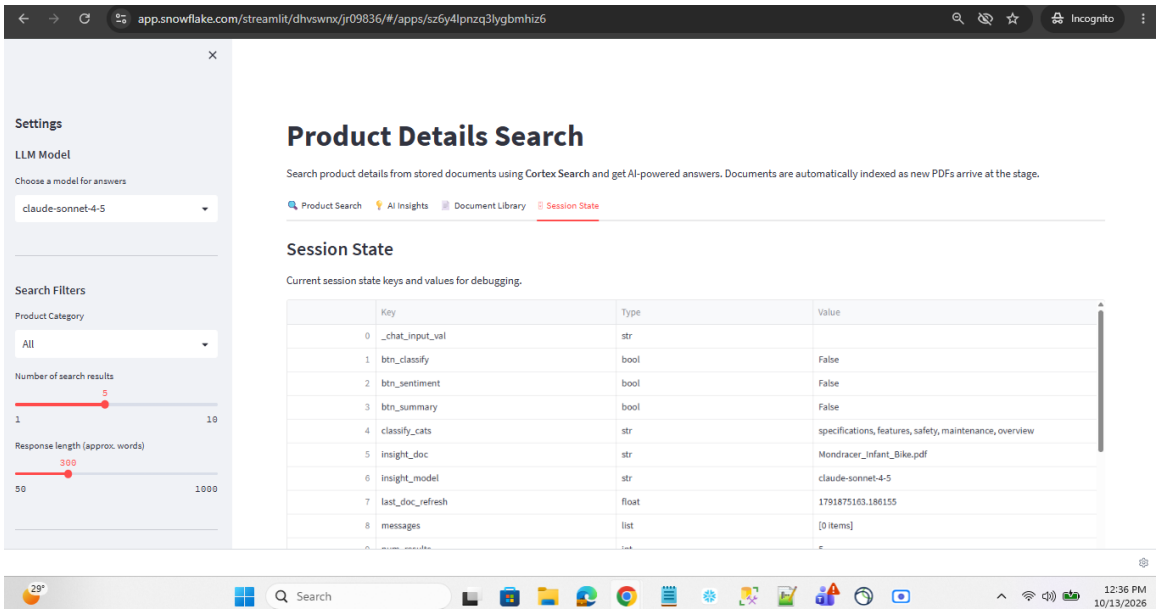
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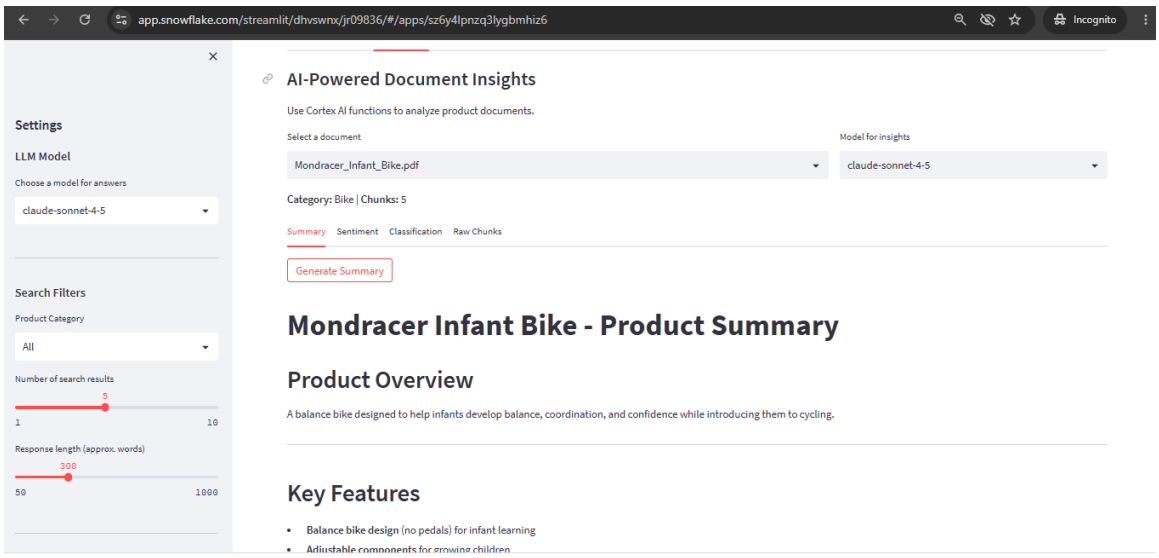
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5. Evaluation Metrics

Recommended evaluation metrics include Retrieval Precision, Context Relevance Score, Answer Faithfulness, Answer Relevancy, Response Latency, Hallucination Rate, User Satisfaction Score, and Citation Accuracy.



Also, App should handle all the below selection criterion on the UI before generating any of the response :

LLM Model => option to select the different models to get the response

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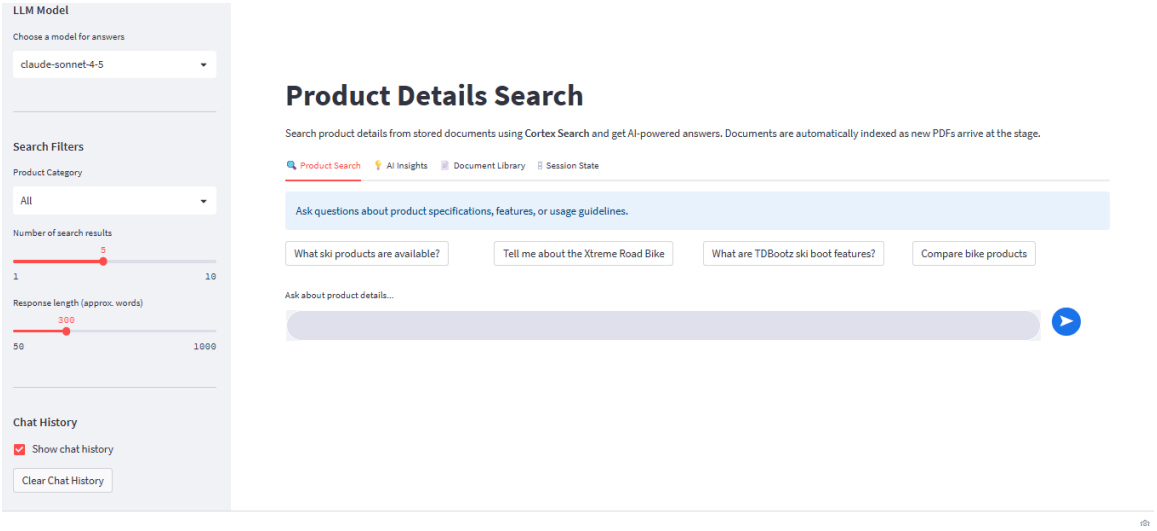
Product Category => option to select the documents of specific type or All in order the generate the response

Number of Search Results => 1-10

Response Length => 50-1000

Chat History => option the select whether chat history is needed to show on UI or not within the session . By default show chat history is on.

Clear Chat History=> Button to clear the chat history from the UI



6. Inferences

The RAG architecture significantly improves answer accuracy by grounding LLM responses in enterprise documents. Snowflake Cortex Search provides scalable retrieval while Streamlit enables rapid development of business-friendly interfaces. The solution reduces hallucinations and provides trustworthy responses with traceable document context.

Component	Technology	Purpose
UI	Streamlit	User interaction and visualization
Retrieval	Snowflake Cortex Search	Semantic search across indexed PDFs
LLM	Cortex AI Models	Answer generation and insights

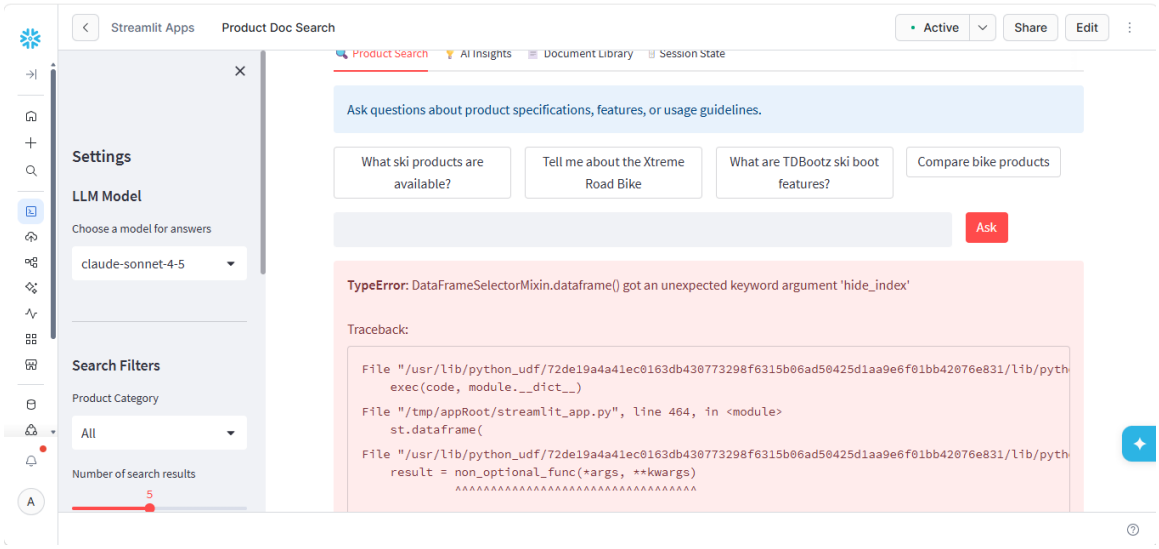
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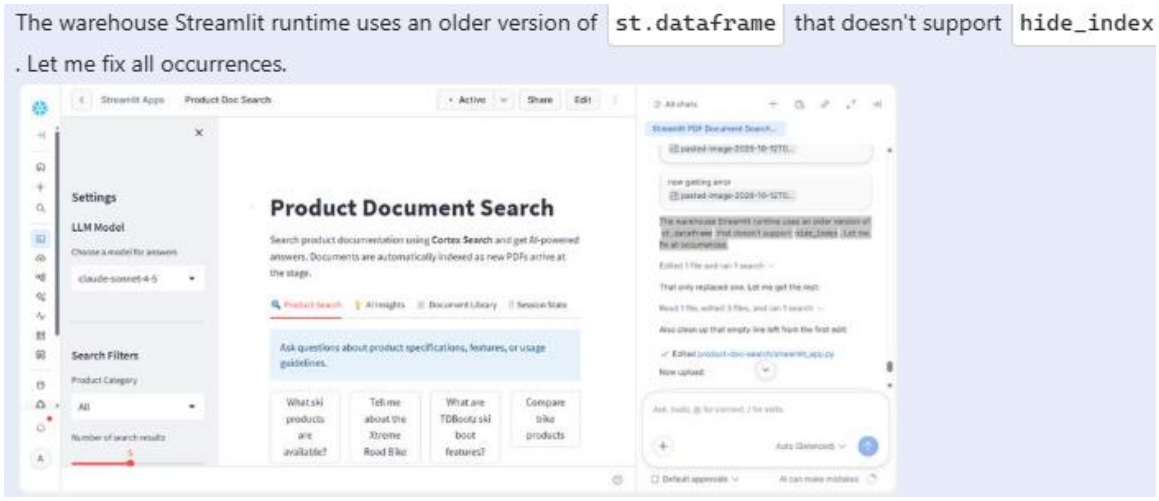
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Storage	Snowflake Stage/Tables	Document repovsitory

7. Challenges

A>

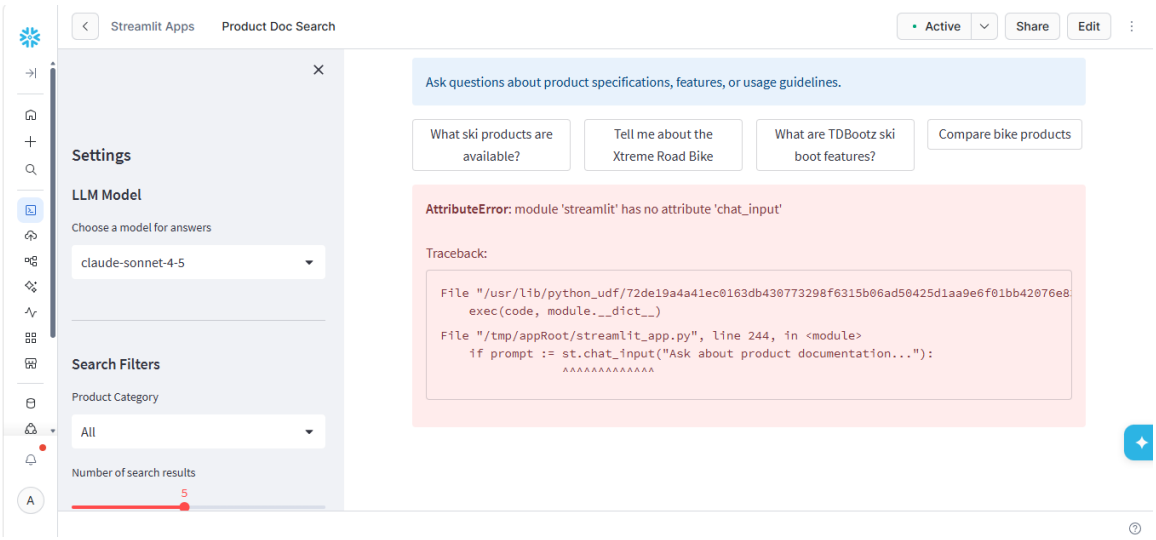


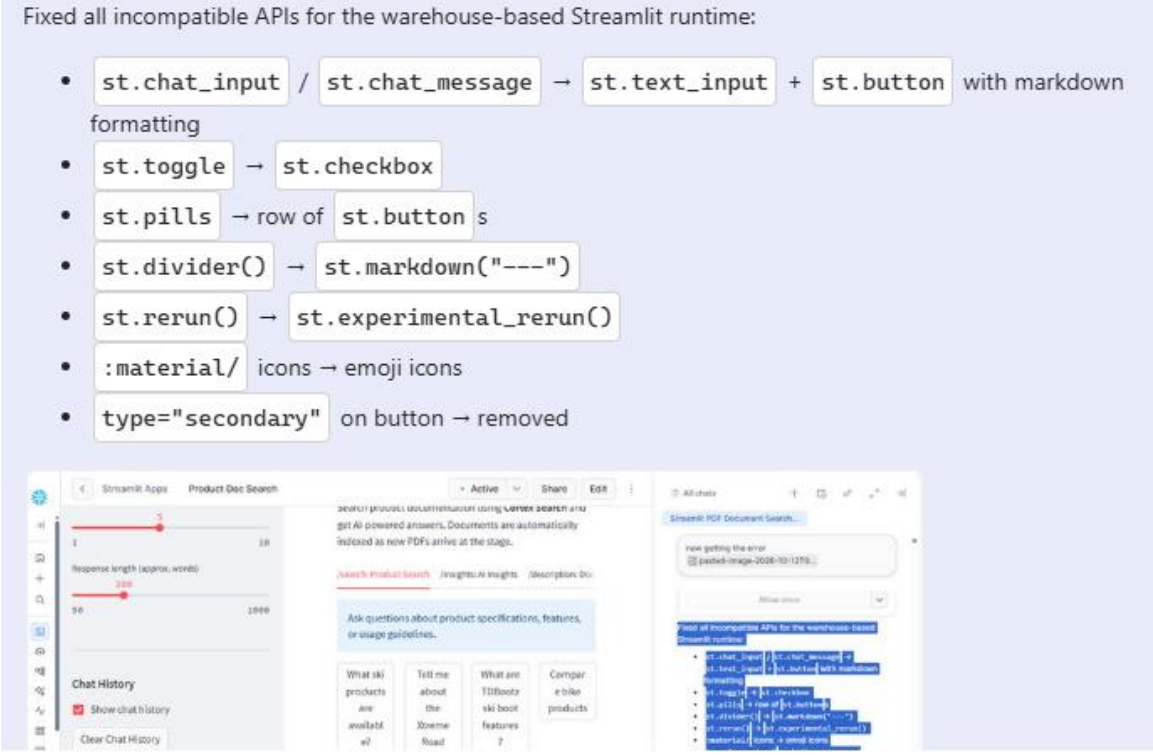


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snowflake.snowpark.exceptions.SnowparkSQLException: (1304): 252004: Binding parameters must be a list: {'svc': 'CC_QUICKSTART_CORTEX_SEARCH_DOCS.DATA.CC_SEARCH_SERVICE_CS', 'body': '{"query": "What ski products are available?", "columns": ["chunk", "relative_path", "category"], "filter": {}, "limit": 5}'}

Traceback:

```
File "/opt/streamlit-runtime/streamlit_app.py", line 252, in <module>
    search_results = search_docs(
                     ^^^^^^^^^^^^

File "/opt/streamlit-runtime/streamlit_app.py", line 97, in search_docs
    ).collect()
      ^^^^^^^^^
```

The `snowflake.core` package isn't available in the Streamlit runtime. I'll switch the Cortex Search call to use the `SEARCH_PREVIEW` SQL function instead.

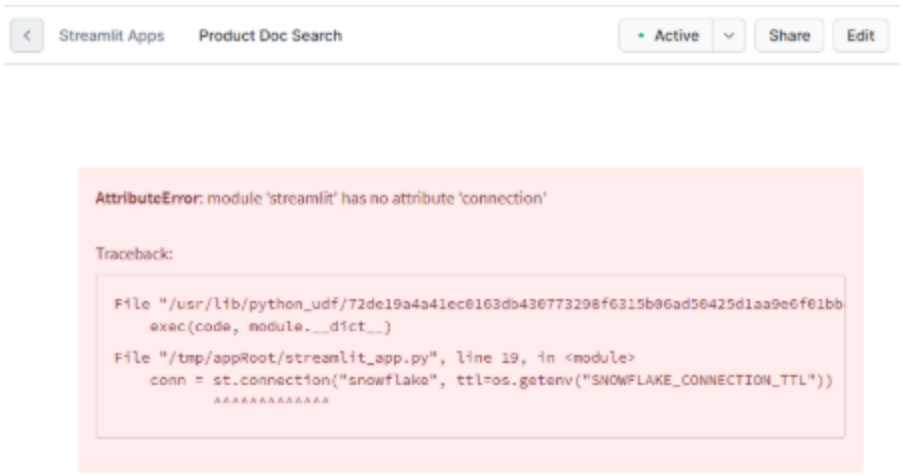
Read 1 file and edited 1 file

Fixed. The `search_docs` function now uses `SNOWFLAKE.CORTEX.SEARCH_PREVIEW` (a SQL function) instead of `snowflake.core.Root` (Python SDK not available in the Streamlit runtime). Click **Run** again to reload the app.

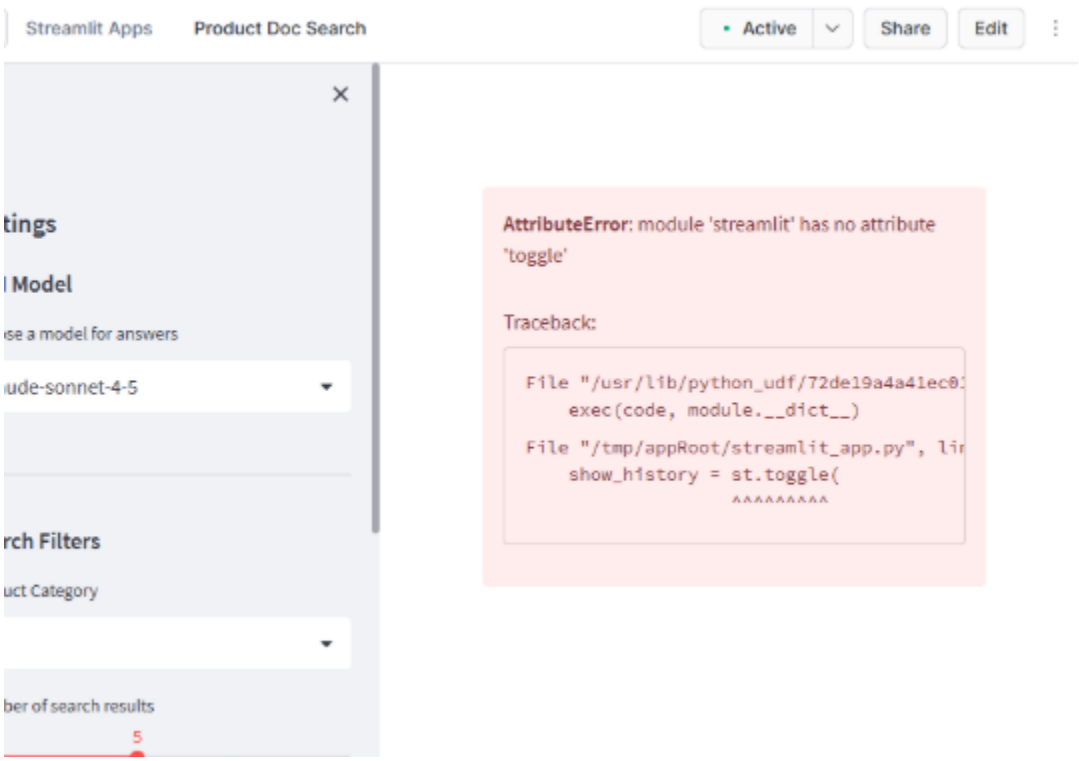
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